

Playwright Intermittent Failure Root Cause Investigation Report

Scenario: 100 Playwright test cases with 30% intermittent failures.
Prepared for engineering investigation.

1. Executive Summary

This report evaluates competing hypotheses without prematurely assigning a root cause. Current evidence most strongly supports a synchronization-related failure pattern, while environmental differences remain a strong competing explanation. Locator and data issues are weakly supported by the available evidence.

2. Problem Statement

Metric	Observation
Total Tests	100
Intermittent Failure	30%
Failure Distribution	70% after navigation; 20% table rendering; 10% random
CI Failure Rate	35%
Local Failure Rate	4%
Trace	Navigation finished; API sometimes pending; DOM rendered twice; locator

3. Competing Hypotheses

Hypothesis	Supporting Evidence	Weakening Evidence	Confidence
Synchronization	Post-navigation failures, pending API updates	DOM rendered before application ready	High
Environment	35% CI vs 4% local	Does not alone explain render sequence	Medium
Application Defect	Double render may indicate UI issue	Manual reproduction yet	Low
Locator	None significant	Locator resolves successfully	Very Low
Data	None	Database identical	Negligible

Important: These confidence levels are investigative judgments, not statistical probabilities.

4. Evidence Assessment

Evidence A

70% of failures occur after navigation.

Assessment: Strongly increases synchronization confidence.

Evidence B

20% occur during table rendering.

Assessment: Consistent with asynchronous rendering.

Evidence C

CI failure rate is much higher than local.

Assessment: Strong evidence for environment as a contributing factor.

Evidence D

API request remains pending after navigation.

Assessment: Supports timing-related investigation.

Evidence E

Table DOM appears twice.

Assessment: Suggests framework re-render or lifecycle update.

Evidence F

Locator resolves successfully.

Assessment: Weakens locator hypothesis.

Evidence G

Database data identical.

Assessment: Strongly weakens data hypothesis.

5. Investigation Strategy

Prioritize verification steps by expected information gain rather than implementation effort.

Priority	Verification Step
1	Compare pass vs fail Playwright traces.
2	Measure network timing in CI versus local.
3	Attempt manual reproduction.
4	Inspect DOM lifecycle to determine whether the first table instance is replaced.
5	Capture CI CPU, memory, and worker utilization.
6	Correlate browser console, network logs, and screenshots for every failure.

6. Decision Framework

Decision Tree Failure ↓ Does rerun pass? ─ Yes → Investigate synchronization and timing. ─ No ↓ Manual reproduction? ─ Fails → Application defect becomes stronger.
─ Passes → Investigate environment, automation timing, and rendering lifecycle.

7. Conclusion

Based on currently available evidence, synchronization is the leading hypothesis. However, the data do not justify declaring it the confirmed root cause. The investigation should continue by collecting higher-value evidence that can differentiate synchronization, environment, and application behavior with minimal ambiguity.